Faithful augmentation at a symmetric wing

Human Compression project: a component for L019

Retain the actual perfect-split setting of the acquired theorem *Symmetric wings at a perfect split*. Thus a transitive string C-marking $(\rho_0, \dots, \rho_{r-1})$ acts on $A \sqcup B$, its left marks are supported on A, its right marks on B, and

$$q = \rho_i = (a\ b)\alpha\beta, \quad a \in A, \quad b \in B,$$

with α supported on $A \setminus \{a\}$ and β on $B \setminus \{b\}$. The omission of q has exactly the two orbits A, B. Write

$$L = \langle \rho_0, \dots, \rho_{i-1} \rangle, \quad R = \langle \rho_{i+1}, \dots, \rho_{r-1} \rangle, \quad H = \langle L, q \rangle, \quad J = \langle q, R \rangle.$$

The groups L, R are actual transitive groups on their supports. The intersection property gives $L \cap J = 1$ and $H \cap J = \langle q \rangle$. We use neither a general quotient intersection property nor a converse to the endpoint central-twist construction.

Lemma 1. *Suppose $L = \mathrm{Sym}(A)$ and α, β are both nonidentity. Restriction of J to $\{a\} \cup B$ is faithful. Its image*

$$K = \langle q', R \rangle, \qquad q' = (a\ b)\beta,$$

is a doubly transitive string C-group. It admits a surjective homomorphism $\chi : K \longrightarrow C_2$ which kills R and sends q' to the nonidentity element. Moreover

$$H = \mathrm{Sym}(A \cup \{b\}) \times \langle \beta \rangle, \qquad K \neq \mathrm{Alt}(\{a\} \cup B), \mathrm{Sym}(\{a\} \cup B).$$

Proof. Nontriviality of the internal factors implies $|A|, |B| \geq 3$. Since $\alpha \in L$, the group H contains $q\alpha = (a\ b)\beta$. Choose $c \in A \setminus \{a\}$. Then $(a\ c) \in L$, and

$$\big((q\alpha)(a\ c)\big)^3 = \beta :$$

the factors on $\{a, b, c\}$ and the support of β are disjoint, and the former is a three-cycle. Thus H contains β and $(a\ b)$. Together with L, this transposition generates $\mathrm{Sym}(A \cup \{b\})$. The reverse inclusion follows from the displayed supports of the generators, proving the direct-product formula.

On $A \setminus \{a\}$ every element of J acts as either 1 or α. Consequently the kernel of its restriction to $\{a\} \cup B$ is contained in $\langle \alpha \rangle \leq L$. Since $L \cap J = 1$, that kernel is trivial. The actual submarking of J therefore transfers by an isomorphism to the displayed marking of K, preserving every subgroup intersection and every nonidentity generator. The group K is transitive, and its point stabilizer at a contains the transitive group R on the complement B. Hence it is doubly transitive and, in particular, primitive.

Use the inverse of this restriction isomorphism, followed by restriction to $A \setminus \{a\}$. This defines the stated homomorphism χ onto $\langle \alpha \rangle \cong C_2$. An alternating group has no nontrivial homomorphism to C_2, since it is generated by three-cycles. This excludes that image. If K were the full symmetric group, its nontrivial character χ would be the sign character. The condition $\chi(q') \neq 1$ would then make β even, because $q' = (a\ b)\beta$. Thus the unique lift to J of the permutation $\beta \in K$ would act trivially on $A \setminus \{a\}$ and would be precisely the supported permutation β. The direct-product formula would place this nonidentity element in $H \cap J = \langle q \rangle$. But β fixes a while q moves a, a contradiction. □

Theorem 2. *Suppose $2r > |A| + |B| + 2$, $L = \mathrm{Sym}(A)$, and both internal factors are nontrivial. Then either*

$$2i > |A| + 2, \tag{1}$$

or $|B| = 5$, R has rank two and order ten, $|A|$ is even, and $i = |A|/2 + 1$. In the latter case the faithful image K from Lemma 1 is the degree-six copy of S_5, and its ordered marking, with q' first, is conjugate to one of the following two tuples:

$$\begin{aligned} &\big((1\ 2)(3\ 4)(5\ 6),\ (2\ 3)(4\ 5),\ (3\ 5)(4\ 6)\big),\\ &\big((1\ 2)(3\ 6)(4\ 5),\ (2\ 3)(4\ 5),\ (3\ 5)(4\ 6)\big). \end{aligned} \tag{2}$$

Their types are respectively $[6, 5]$ and $[4, 5]$, and their attachment point is $a = 1$, with $b = 2$.

Proof. Put $t = r - i - 1$, $m = |B| + 1$ and $d = t + 1 = \mathrm{rank}(K)$. The exact acquired primitive theorem of L047 applies to K: Lemma 1 supplied its faithful primitive C-action and excluded both natural alternating and symmetric groups. If $d < m/2$, integrality gives $2t \le |B| - 2$. Consequently

$$2i = 2r - 2 - 2t > |A| + |B| - 2t \ge |A| + 2,$$

as required.

The complete primitive exceptions with $d \ge m/2$ are the retained degree-six A_5, S_5 tuples and the degree-ten rank-five S_6 tuple. We test their twelve exact ordered census representatives in both orientations. For each tuple we test every possible anchor, transitivity of its tail on the complement, and the existence of the character which sends the first mark to $1 \in \mathbb{F}_2$ and the others to zero. Only four oriented records survive: source indices $(3, 1), (4, 1), (6, 0), (8, 0)$, where the second coordinate specifies reversal and indices start at zero. The first and third are identical and give the first tuple in (2). The second is the second displayed tuple. The fourth is its conjugate under the point map $(1, 2, 3, 4, 5, 6) \mapsto (1, 2, 4, 5, 6, 3)$.

For clarity, the character test is exact: during finite group closure attach to each visited permutation the sum modulo two of its first-generator occurrences. A conflict excludes the character; if no conflict occurs, every multiplication by every generator respects the assigned values, which defines the required homomorphism. Endpoint anchors and complementary orbits are finite point tests. The independent certificate retains every tested tuple, the character results and the explicit conjugacies. Its completeness premise is the acquired twelve-tuple classification, not this filtering calculation.

In either surviving case $|B| = 5$ and $t = 2$. The assumed rank inequality becomes $2i > |A|+1$. If (1) fails, the even integer $2i$ must equal $|A|+2$. This forces $|A|$ to be even and gives the stated residual value of i. The tail in both tuples has order ten and is transitive on the five points other than 1. □

Lemma 3. *Suppose the actual left wing has a perfect split at an interior index j, with $1 \le j \le i - 2$. Then it is also a perfect split of the full marking. If one of its internal factors was trivial in the wing, that factor remains trivial in the full marking.*

Proof. Write $A = C \sqcup D$ for this wing split, whose crossing pair is $(c, d) \in C \times D$. The marks before j generate a transitive group P supported on C, and those strictly between j and i act transitively on D. The former list is nonempty, so $|C| > 1$. The split mark $q = \rho_i$ commutes with every member of P; as C is its only nontrivial orbit, q preserves C. It also commutes with ρ_j. The latter has exactly one edge from C to its complement, namely (c, d). Therefore q fixes both c and d. An element centralizing a transitive group and fixing one point fixes the entire orbit, so q fixes C pointwise.

The old attachment point a lies in D: the mark ρ_{i-1} moves a and is supported there, since $j < i - 1$. It follows that the right side of the new omission is transitive on $D \cup B$, using the old transitive action on D, the crossing (a, b) and the transitive action of R on B. All right-side

marks fix C. Thus $C, D \cup B$ are precisely the required two omission orbits, and ρ_j still has its single crossing edge and the required supported factors. Those factors have not changed, proving the final assertion. □

Role in the recovery and attribution. The separately reviewed actual-wing theorem ensures that at least one wing is symmetric in the stated high-rank range; reverse the marking to call it L. The present component then supplies a legitimate restriction isomorphism and the complete two-type boundary to descending inside that symmetric wing. The rank theorem and finite primitive classification are the historical results retained through the complete L047 recovery. The character and intersection arguments above are new proof interfaces in this recovery. Lemma 3 is the actual support argument underlying the corresponding inheritance step in the source; no priority for that elementary observation is claimed.

The residual even-degree case of Theorem 2 is not excluded here. Nor does this note prove the existence of an interior perfect split in a smaller wing, verify every required leaf-kernel condition, or establish the stability bijection. It supplies no whole-original, open-question or Lean credit.